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RESULTS

Results

01 · KRUSKAL-WALLIS

nonparametric 3+ group comparison

Table 1. Rank summary

Group	N	Median	IQR	Mean rank
Blended	80	5.00	3.00	125.96
Flipped	80	2.00	4.00	72.51
Lecture	80	6.00	3.00	163.03

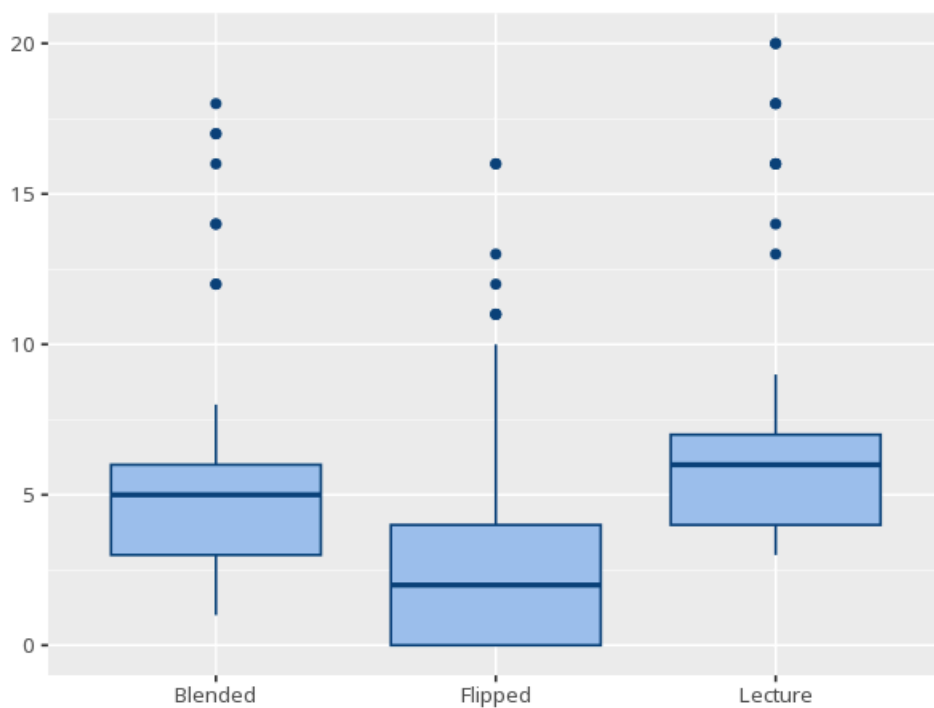
Table 2. Kruskal-Wallis test

H	df	p	ϵ^2 [95% CI]
69.55	2	<.001	0.29 [0.22, 1.00]

Table 3. Dunn post-hoc

Pair	Z	p_{adj}
Blended - Flipped	-4.90	<.001
Blended - Lecture	3.40	<.001
Flipped - Lecture	8.29	<.001

Figure. Distribution by group



Understanding the numbers

df. The degrees of freedom for the Kruskal-Wallis test, based on the number of groups. Here, $df = 2$.

ϵ^2 . The effect size for the Kruskal-Wallis test - the proportion of rank variance explained by group membership. Here, $\epsilon^2 = 0.29$ [0.22, 1.00].

H. The Kruskal-Wallis test statistic, testing whether any group's ranks differ systematically from the others. Here, $H(2) = 69.55$.

IQR. The middle 50% spread of scores within each group (descriptive, not part of the test itself). Here, the IQR ranges from 3.00 in Blended to 4.00 in Flipped.

Mean rank. The average rank of each group's scores when all scores are ranked together - the quantity H is calculated from. Here, the mean rank ranges from 72.51 in Flipped to 163.03 in Lecture.

Median. The middle value of each group's scores. Here, the median ranges from 2.00 in Flipped to 6.00 in Lecture.

N. The total number of observations used across all groups. Here, $N = 240$.

p. The probability of rank differences this large (or larger) across groups if all groups truly had the same distribution. Here, $p < .001$ - below your significance threshold, at least one group differs.

p adj. The Dunn post-hoc adjusted p-value for a specific pair of groups. Here, the strongest pairwise difference is Flipped - Lecture, $p \text{ adj} < .001$.

Z. The Dunn post-hoc test statistic comparing a specific pair of groups' mean ranks. Here, for Flipped - Lecture, $Z = 8.29$.

How to read this test

The nonparametric counterpart to one-way ANOVA. The H/p tests whether any group tends to have systematically higher or lower values (stochastic dominance) - read it as a "median difference" only when the groups have similar distribution shapes. If significant, Dunn's post-hoc shows which pairs differ, with adjusted p-values. ϵ^2 is the effect size. Your significance threshold (α) is 0.05.

APA template: A Kruskal-Wallis test gave $H(2)=69.55$, $p < .001$, $\epsilon^2=.29$ [.22, 1.00].

Statistical basis: Kruskal-Wallis H test (nonparametric counterpart to one-way ANOVA) (Kruskal, W. H., Wallis, W. A. (1952). "Use of ranks in one-criterion variance analysis." Journal of the American Statistical Association, 47(260), 583-621.); Dunn's post-hoc pairwise comparisons (Kassambara A (2023). rstatix: Pipe-Friendly Framework for Basic Statistical Tests. R package.); ϵ^2 effect size (Ben-Shachar MS, Lüdtke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." Journal of Open Source Software, 5(56), 2815.). Full references in the exported CITATIONS.txt.

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